

PROJECT BRIEF

AI-Powered Digital Twin for a Low Carbon Intelligent Building (LCIB)

CNN-GAN Hybrid Deep Learning | Edge Computing | Real-Time Inference

Project Lead	Prof. Mohammad Saud Afzal
Affiliation	IIT Kharagpur, Civil Engineering
Duration	12 months
Total Effort	850 working hours
Date	April 2025

For potential collaborators and project staff — Confidential

1. What This Project Is About

We have an opportunity to develop a full-scale, production-ready AI digital twin for a Low Carbon Intelligent Building (LCIB) at a European engineering institution. I am looking for collaborators and project staff who can contribute to specific technical streams within this programme.

The core idea: we build a deep learning system — specifically, a CNN-GAN hybrid architecture — that learns from the building's sensor network and can simulate how the building behaves under real and hypothetical conditions. The twin runs on edge hardware inside the building for real-time inference (sub-500ms response), and it is calibrated to 95% or better accuracy against a physics-based numerical model built in EnergyPlus.

The facility includes an existing instrumented cold storage room that serves as our Phase 1 training and validation environment. Cold storage rooms are excellent for this purpose — tightly controlled parameters, clean sensor data, predictable compressor cycling. Published work on comparable setups has achieved temperature prediction errors below 0.25°C at one-minute time steps. That room lets us train and validate models in a controlled space before scaling to the full building sensor network.

The project runs for **12 months** with a total budget of **850 working hours** distributed across six phases. I am sharing this brief to identify people who can take on specific technical roles within that structure.

2. Technical Scope

The work is organised into five technical streams. They overlap — data engineering continues throughout, model development runs in parallel with edge integration from the midpoint. Here is what each stream involves.

2.1 Data Infrastructure and Numerical Twin Foundation

This is the base layer. Using building geometry documentation and sensor data provided by the institution, we construct a physics-based numerical model in EnergyPlus (or equivalent). This numerical twin is the accuracy benchmark for the entire project — it represents what the physics predicts, and we calibrate the AI twin until its outputs are within 95% agreement.

Alongside the numerical model, we design and implement real-time data ingestion pipelines connecting the building's sensor network to the twin's data layer. Protocols involved: BACnet, Modbus, and MQTT. A data quality framework goes in from the start — handling missing readings, sensor drift, and timestamp irregularities. This is not optional work. Every ML model downstream is only as good as the data feeding it.

Skills needed: EnergyPlus modelling or equivalent building energy simulation experience; IoT data pipeline design; familiarity with BACnet/Modbus/MQTT; data quality and cleaning frameworks.

2.2 Machine Learning Data Engineering

Raw sensor streams do not train themselves. Before the deep learning models see anything, the data needs to be transformed into structured, labelled training datasets. This covers feature engineering — figuring out which combinations of sensor readings are most predictive of each target variable — plus the preparation of training/test/validation splits, cross-validation design, and synthetic data augmentation where sensor coverage has gaps.

The cold storage room dataset is the primary training source initially. Building-wide sensor data gets integrated progressively as it becomes available and passes quality checks. Everything is version-controlled and documented so that any training run can be reproduced exactly.

Skills needed: Feature engineering for time-series and spatial sensor data; experience with ML data pipelines; dataset versioning (DVC or similar); familiarity with data augmentation techniques.

2.3 CNN-GAN Deep Learning Model Development

This is the core of the project and the most technically demanding stream, accounting for 250 of the 850 hours.

CNN Component

Convolutional Neural Networks process the spatially distributed sensor readings — a grid of temperature sensors across a floor plan, for instance — as 2D feature maps. The CNN extracts spatial patterns: thermal gradients, hotspots, energy distribution profiles that time-series models alone would miss. The same architecture handles structural sensor data (accelerometers, strain gauges) for vibration signature extraction if such sensors are present.

GAN Component

The Generative Adversarial Network is what gives the twin its simulation capability — the ability to generate building behaviour under conditions never physically observed. Two competing networks: a Generator producing synthetic building state sequences (temperature profiles, energy demand, humidity) and a Discriminator trained to distinguish real from generated. Adversarial training pushes the Generator towards increasingly realistic outputs. We use a Wasserstein GAN with gradient penalty (WGAN-GP), which is the current standard for stable training on time-series data and avoids mode collapse.

In practical terms, this allows end-users to pose scenario questions: what happens to energy consumption if glazing changes, or occupancy doubles, or HVAC setpoints shift by 2°C? The twin generates plausible simulated responses without any physical modification to the building.

LSTM Integration

A Bidirectional LSTM handles the sequential data streams — energy demand over time, CO₂ accumulation curves, HVAC load cycles. Its outputs combine with the CNN's spatial features as joint conditioning input to the GAN. A multi-head self-attention mechanism focuses the model on the most relevant time steps per prediction.

Skills needed: Deep learning model design and training (TensorFlow 2.x or PyTorch 2.x); GAN architectures, ideally WGAN-GP; CNN and LSTM implementation; hyperparameter optimisation (Bayesian); experiment tracking (MLflow); mixed-precision training on GPU.

2.4 Edge Computing Integration

We deploy AI inference onto edge hardware physically inside the building, rather than relying on cloud round-trips. This eliminates communication latency and gets anomaly detection and control responses under 500 milliseconds — fast enough for actual HVAC control and safety monitoring.

The work involves configuring edge hardware (NVIDIA Jetson Xavier NX or industrial equivalent), deploying optimised model versions using quantisation and pruning, and setting up middleware to route data between sensors, edge nodes, and the central server. The edge nodes must operate autonomously during network outages — continuous monitoring cannot depend on server uptime.

Skills needed: Edge AI deployment (Jetson or similar); model quantisation and pruning (TensorRT, ONNX); IoT middleware and fog computing; real-time systems; latency benchmarking.

2.5 Calibration, Validation, and Dashboard

Hitting 95%+ accuracy requires iterative calibration of the AI twin against the numerical twin. We use Bayesian optimisation to tune both AI hyperparameters and numerical model inputs — thermal mass, infiltration coefficients, HVAC performance curves — until convergence. Primary metrics: $R^2 \geq 0.95$ and $RMSE \leq 0.5^\circ\text{C}$ for temperature.

SHAP (SHapley Additive exPlanations) analysis is applied to all model outputs so that predictions are interpretable and auditable. This is not a nice-to-have — it is essential for institutional trust and for any review or reporting requirements.

The executive dashboard is a browser-based web application: React.js front-end, Python FastAPI back-end, time-series database (InfluxDB or PostgreSQL), containerised with Docker. It provides real-time building visualisation, energy demand forecasts, scenario modelling, anomaly alerts with root-cause attribution, and automated energy/carbon reporting.

Skills needed: Bayesian optimisation and model calibration; SHAP/XAI methods; React.js and Python FastAPI for dashboard development; time-series database experience; Docker.

3. Architecture Summary

3.1 Deep Learning Stack

- **Input modalities:** 2D spatial sensor grids (CNN), sequential time-series (LSTM), scalar conditioning vectors (scenario parameters, weather, occupancy).
- **CNN backbone:** 5 convolutional layers, ResNet skip connections, batch normalisation, ReLU activations.
- **GAN formulation:** WGAN-GP. Generator: 6 transposed convolutional layers with fully connected conditioning. Discriminator: 5 convolutional layers with spectral normalisation.
- **LSTM module:** 2-layer Bidirectional LSTM, 128 hidden units, multi-head self-attention.
- **Framework:** TensorFlow 2.x or PyTorch 2.x (to be confirmed). Experiment tracking via MLflow.

- **Training:** Mixed-precision (FP16/FP32) on GPU. A physics-informed constraint layer may be added to enforce thermodynamic plausibility if convergence requires it.

3.2 Edge Computing Stack

- **Device tier:** IoT sensors, smart meters, HVAC actuators via BACnet/IP, Modbus TCP, MQTT over TLS.
- **Edge tier:** Edge AI node(s) running optimised inference models. Target: <500ms anomaly detection and control response.
- **Server tier:** Full model training, calibration, scenario generation, and dashboard hosting. Edge nodes synchronise periodically for model updates and telemetry aggregation.

3.3 Dashboard Stack

- **Front-end:** React.js with real-time floor plan overlays, scenario modelling interface, anomaly alert log.
- **Back-end:** Python FastAPI.
- **Database:** InfluxDB or PostgreSQL for time-series storage.
- **Deployment:** Docker containers. Role-based access (4 levels).

4. Accuracy and Calibration Targets

These are the quantitative benchmarks the project must meet:

Metric	Target
R ² (primary output variables)	≥ 0.95
RMSE (temperature)	≤ 0.5°C
MAPE (energy demand)	≤ 5%
MAE	Within 1 standard deviation of sensor noise floor
Cold storage spatial variation	±0.5°C of setpoint under AI-guided control
Edge inference latency	< 500 ms

Calibration uses multi-stage Bayesian optimisation assisted by particle swarm optimisation, iterating at one-minute time steps. If the 0.95 target proves difficult within the calibration budget, $R^2 \geq 0.90$ is the documented fallback. Any shortfall is reported transparently.

5. Effort Breakdown — 850 Hours

The table below shows how the 850 hours are distributed. Phase 3 (model development) is the largest single block, but note that data infrastructure and engineering together account for about 30% of total effort. That is deliberate — a sophisticated model trained on poor data will not hit the targets.

Phase	Key Tasks	Hours
Phase 1 — Data & Foundation (M1–M2)	Building geometry and sensor data review; EnergyPlus numerical baseline; data ingestion pipelines (BACnet, Modbus, MQTT); cold storage data audit and cleaning; quality framework.	140
Phase 2 — ML Data Engineering (M2–M4)	Feature engineering; cold storage training dataset preparation; train/test/validation splits; data augmentation; documentation for reproducibility.	120
Phase 3 — CNN-GAN Development (M3–M7)	CNN design and training; WGAN-GP adversarial training; hybrid integration; LSTM with attention; Bayesian hyperparameter optimisation; MLflow experiment tracking.	250
Phase 4 — Edge Computing (M6–M9)	Edge hardware configuration; model quantisation/pruning; fog middleware; real-time inference testing (<500ms target); anomaly detection deployment.	120
Phase 5 — Calibration & Validation (M8–M11)	Bayesian calibration against numerical twin; R^2 , RMSE, MAE, MAPE; SHAP analysis; confidence intervals; stress-testing under peak and fault conditions.	130
Phase 6 — Dashboard & Handover (M10–M12)	React/FastAPI dashboard; scenario comparison module; alert system; final documentation package.	90
TOTAL		850

Note: Phases 3 and 4 overlap during Months 6–7, allowing early integration testing of CNN-GAN output with the edge layer.

6. Project Timeline

The 12-month clock starts once the institution delivers a complete building data package. Phase boundaries shown below are indicative.

Phase	Q1 (M1–M3)	Q2 (M4–M6)	M7–M8	M9	Q4 (M10–M12)
Phase 1 — Data & Foundation	■				
Phase 2 — ML Data Engineering	■	■			
Phase 3 — CNN-GAN Development	■	■	■		
Phase 4 — Edge Computing		■	■		
Phase 5 — Calibration & Validation			■	■	
Phase 6 — Dashboard & Handover				■	■

7. Cold Storage Room — Data Specifications

The cold storage room is the Phase 1 training environment. The quality and completeness of its data is the single most important factor in how quickly the AI models converge. Below is the sensor data specification for model training.

Parameter	Sensor / Source	Min. Frequency	Priority
Internal air temperature (°C)	Calibrated T/RH — min. 3 locations	1 min	Critical
Relative humidity (%RH)	Calibrated T/RH — min. 3 locations	1 min	Critical
Compressor energy (kWh)	Smart meter on compressor circuit	1 min	Critical
Compressor ON/OFF	Digital I/O from BMS or relay logger	Event-based	Critical
Condenser/evaporator temp	PT100 surface-mount	5 min	High
Door open/close events	Door contact sensor	Event-based	High
External ambient temp	Weather station or BMS feed	5 min	High
CO ₂ concentration (ppm)	CO ₂ sensor (e.g. Senseair S8)	5 min	Medium
Refrigerant pressure (bar)	Pressure transducer — HP/LP lines	5 min	Medium
Fan speed / airflow	VFD data or tachometer	5 min	Medium
Lighting energy (kWh)	Smart meter sub-circuit	15 min	Low
Product load / stored mass (kg)	Manual log or load cell	Daily	Low

We also need: architectural drawings (CAD or PDF), thermal specifications (U-values, insulation), refrigeration system documentation (compressor datasheet, COP, refrigerant type), at least three months of continuous sensor data at one-minute intervals, and historical energy consumption records.

8. Expected Outcomes

When the twin is operational, the platform supports:

Predictive Energy Management

24–48 hour energy demand forecasts enabling HVAC pre-conditioning. Published evidence on deep learning digital twins reports 15–35% annual HVAC energy savings against conventional PID control.

Cold Storage Optimisation

Temperature maintained within $\pm 0.5^{\circ}\text{C}$ of setpoint while minimising compressor runtime. Comparable published studies show ~23% refrigeration energy reduction and 47% improvement in spatial temperature uniformity.

Structural Health Monitoring

Continuous CNN-based vibration and strain analysis (where sensors exist). Anomalies — unusual mode frequencies, localised stress concentrations — flagged in real time for condition-based maintenance.

Scenario Modelling

End-users can test the impact of building modifications, energy system upgrades, climate projections, or occupancy changes through the GAN's generative capability. No physical changes required.

Automated Reporting

Monthly energy and carbon summaries aligned with EU EPBD (2024/1275), exportable as PDF.

Predictive Maintenance

Edge-deployed anomaly detection catches equipment degradation — declining COP, increased infiltration, abnormal fan power — typically weeks before failure.

9. Roles and Expertise Needed

I am looking to fill several specific roles within this project. You do not need to cover all of the below — if your expertise aligns with even one stream, I would like to hear from you. The roles are structured around the technical streams described earlier.

Role	Key Expertise	Primary Phase(s)
Building Energy Modeller	EnergyPlus or equivalent; building physics; thermal simulation; calibration of numerical models against sensor data.	Phase 1, Phase 5
IoT / Data Pipeline Engineer	BACnet, Modbus, MQTT; real-time data ingestion; data quality frameworks; sensor data cleaning and validation.	Phase 1, Phase 2
ML Data Engineer	Feature engineering for sensor data; dataset preparation and versioning; data augmentation; Python data stack.	Phase 2
Deep Learning Researcher / Engineer	CNN, GAN (ideally WGAN-GP), LSTM architectures; TensorFlow or PyTorch; hyperparameter optimisation; MLflow.	Phase 3
Edge AI Engineer	Jetson or equivalent edge platforms; model quantisation (TensorRT, ONNX); fog middleware; real-time inference systems.	Phase 4
Full-Stack Developer (Dashboard)	React.js; Python FastAPI; InfluxDB or PostgreSQL; Docker; data visualisation for real-time systems.	Phase 6

Role	Key Expertise	Primary Phase(s)
Calibration / XAI Specialist	Bayesian optimisation; SHAP analysis; model validation methodology; statistical testing.	Phase 5

Depending on your background and availability, involvement could range from a focused contribution to a single stream (e.g. Phase 3 only) to a sustained role across multiple phases. I am flexible on structure — what matters is that the right expertise is in the room for each phase.

10. Scientific Basis

The technical approach draws on peer-reviewed work published in 2025, including:

- Nature Scientific Reports — systematic review on digital twin-integrated ML frameworks for building energy efficiency via real-time IoT feedback.
- Scientific Reports — attention-based LSTM within a digital twin achieving 83.8% thermal comfort classification accuracy with SHAP explainability.
- ScienceDirect, Energy and Buildings — cold chain digital twin achieving RMSE < 0.25°C at one-minute steps using EnergyPlus with PSO calibration.
- ResearchGate — CFD-ML hybrid study: 23% energy reduction and 47% temperature uniformity improvement in a 2,400 m³ cold storage facility.
- Tandfonline — comprehensive survey confirming CNN-GAN with IoT sensor fusion as current state of the art for building digital twins.
- Frontiers in Built Environment — sustainable innovations in digital twin technology for energy efficiency and indoor environment quality.
- Directive (EU) 2024/1275 — Energy Performance of Buildings Directive.

11. Next Steps

If this project aligns with your expertise and interests, please get in touch. I would be happy to discuss the specific stream(s) where your contribution would fit, the expected time commitment, and the working arrangement. The project timeline is fixed at 12 months, so I am aiming to have the team in place before the data package arrives and the clock starts.

Contact: Prof. Mohammad Saud Afzal, Department of Civil Engineering, IIT Kharagpur.

— End of Document —